

# Title: Effective Elastic Properties of Composite Granular Rocks Using Particulate Discrete Element Modelling

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## **Abstract**

The mechanical behavior of composite granular rocks is a multifaceted phenomenon relevant to various geoenvironmental applications. Traditional homogenization models and continuum mechanics-based numerical methods often fall short of accurately capturing the intricacies of granular materials. Granular materials exhibit heterogeneity and arching mechanisms governing the force networks that ensure system stability. Unlike continuum-based approaches, particulate discrete element methods have an advantage in assessing effective material properties by considering material heterogeneity and grain-level physical interactions. This study evaluates effective elastic properties using particle flow code using flat-joint contact law for composite binary mixtures with a stiff inclusion embedded within a matrix material. We created digital models of granular rocks to simulate the properties of a moderate-strength sandstone matrix. Stiff inclusions were incorporated in two geometric configurations: i) as structural inclusions and ii) as laminated inclusions. Our numerical experiments involved subjecting the models to unconfined compressive, isotropic, and shear stress loading to analyze the material's anisotropic elastic properties. This included the estimation of Young's modulus, Poisson's ratio, bulk modulus, and shear modulus in all directions.

Our findings closely adhere to the Voigt-Reuss and Hashin-Shtrikman bounds within their specific conditions, demonstrating the promising application of the discrete element method in the analysis of composite materials. Additionally, the results adhere to the principles of generalized Hooke's law under small strain conditions. A detailed analysis is carried out on the stress-strain distribution within the specimen, examining its deviation from the idealized concepts such as iso-stress and iso-strain behavior. Additionally, our research provides an in-depth analysis of the distinctive stress-evolution and damage-evolution characteristics exhibited by different geometrical configurations of inclusions. These results offer invaluable insights into the mechanical behavior of composite granular rocks and underscore the potential applications of the discrete element method in addressing rock physics modeling problems encountered in petroleum engineering.